

Determinants of Daily Energy Intake in U.S. Adults

Research Question

How do macronutrient composition and alcohol consumption predict total daily energy intake among U.S. adults?

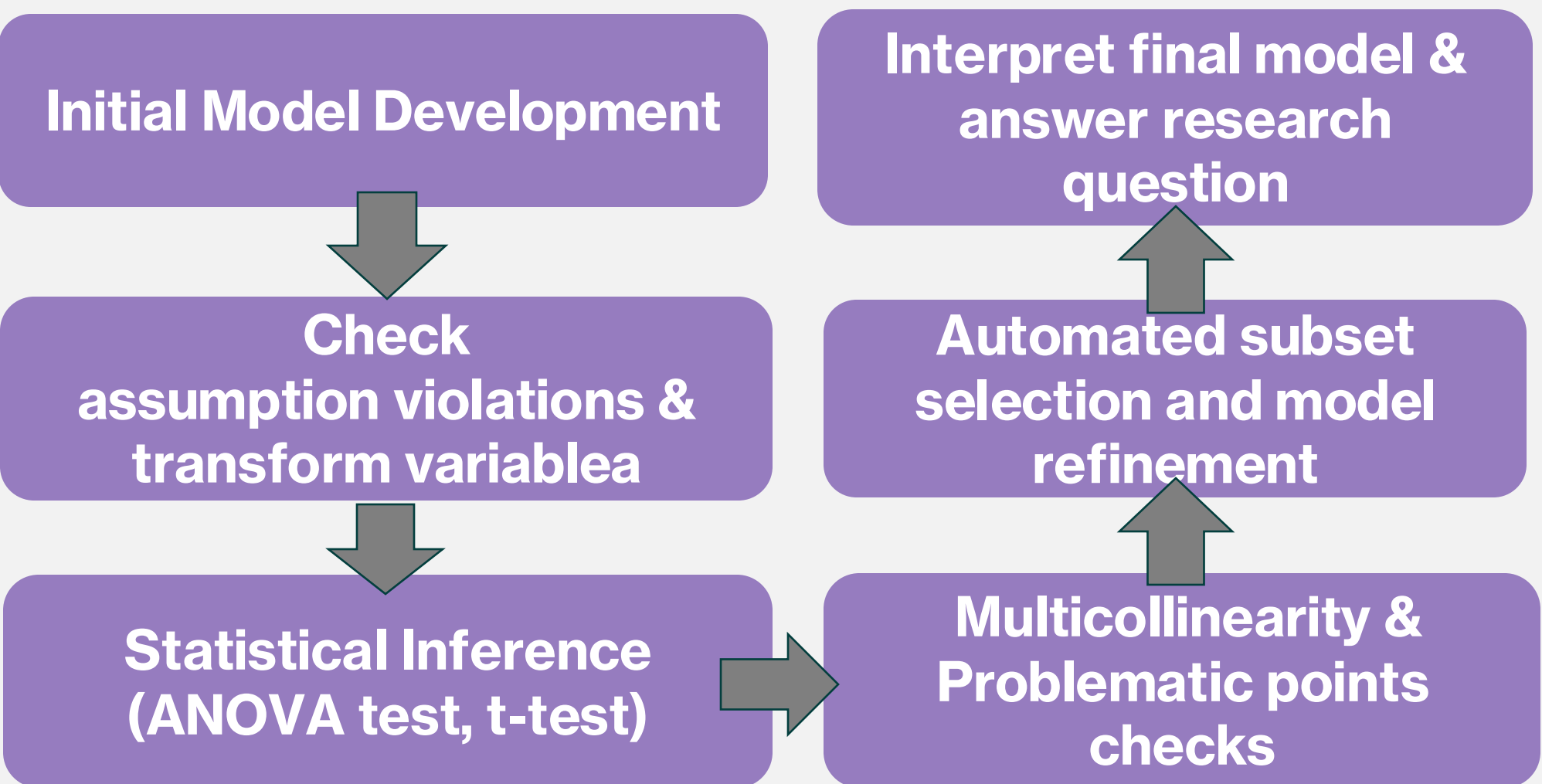
Motivation

- Rising obesity rates highlight the need to understand factors influencing daily energy intake.
- Calorie intake is shaped not only by food quantity, but by the mix of macronutrients and alcohol use.
- By analyzing these components together, this study aims to identify the most important predictors of total daily energy intake in U.S. adults.

Data Collection

This study uses NHANES data, collected through the CDC and USDA's standardized 24-hour dietary recall method. The cleaned dataset contains 8,525 adults with 11 nutrient variables, and right-skewed caloric intake with several high-intake outliers. NHANES' national representativeness, standardized and reliable measurement procedures make it appropriate for modeling determinants of daily energy intake.

Methods of Analysis



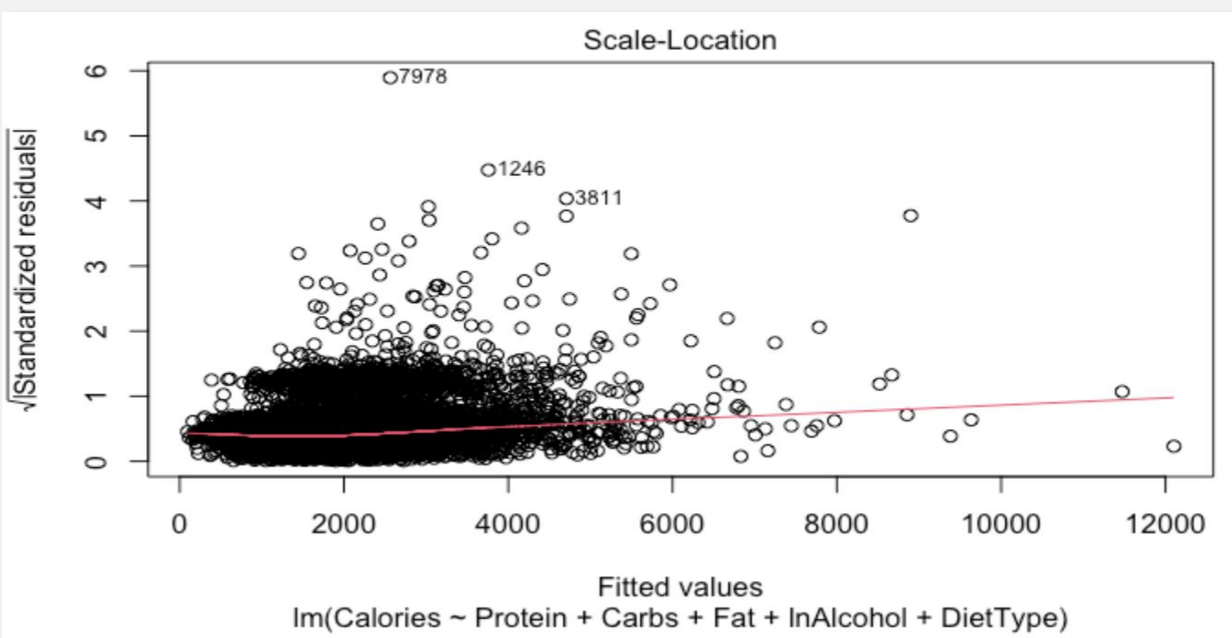
Results

Data Transformation:
Log-transformed Alcohol to fix heteroscedasticity seen in residual plots.

Variable Deletion:
Removed Cholesterol because ANOVA indicated it was not a significant predictor.

Model Diagnostics:
No influential observations were found; ANOVA testing showed significant variables and VIF values showed low multicollinearity

Model refinement:
Automated Subset selection showed best model included Protein, Carbs, Fat, lnAlcohol and DietType



Predictor <chr>	VIF <dbl>
Protein	2.125
Carbs	1.910
Fat	2.710
lnAlcohol	1.027
DietType	1.010

Protein <chr>	Carbs <chr>	Fat <chr>	Sugar <chr>	Fiber <chr>	lnAlcohol <chr>	Sodium <chr>	Potassium <chr>
		*					
	*	*					
*	*	*			*		
*	*	*		*	*		
*	*	*	*	*	*		*
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Final Model

Predictors	Estimates	p
(Intercept)	-38.183	<0.001
Protein	4.261	<0.001
Carbs	3.956	<0.001
Fat	8.77	<0.001
lnAlcohol	99.537	<0.001
DietType (Not on Diet)	19.068	<0.001
R^2 / Adjusted R^2	0.99 / 0.99	<0.001

Conclusion

Protein

Each additional gram predicts +4.26 kcal/day, controlling for all other predictors

Carbs

Each extra gram predicts +3.96 kcal/day, controlling for all other predictors

Fat

Fat has the strongest macronutrient effect; each gram predicts +8.78 kcal/day

lnAlcohol

For every one-unit increase in log(alcohol), intake increases by ~100 kcal/day controlling for all other predictors

DietType

Individuals not on a diet consume ~19kcal/day more than those on a diet, controlling for all predictors

Final model explains ~99% of variation in caloric intake, showing macronutrient composition and alcohol intake almost completely determine daily energy intake.

Limitations

The final model has a few constraints:

- Slight violation of the constant-variance assumption
- Slight violation of normality assumption (deviations in right tail due to unusually large alcohol intake values)
- Problematic observations due to extreme dietary recalls

Although these issues do not significantly affect overall results, further investigation into problematic points and assumptions could increase reliability.

References

Centers for Disease Control and Prevention. (2024, December 16). *About NHANES*. National Health and Nutrition Examination Survey. <https://www.cdc.gov/nchs/nhanes/about/index.html>
NHANES 2013-2014 Dietary Variable List. (2014). Cdc.gov. <https://www.cdc.gov/Nchs/Nhanes/Search/variablelist.aspx?Component=Dietary&CycleBeginYear=2013>